CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 06: AUTOMATED AND EMERGING TECHNOLOGIES

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/12/M/J/24 Q10 (2024 Series)] [Total: 6 marks]

(a) An automated greenhouse maintains optimal temperature. Describe how sensors, a microprocessor, and actuators work together to control the temperature. **[4 marks]**

(b) State two characteristics that define an industrial robot. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) 1. Temperature sensor sends analogue reading to ADC [1 mark]. 2. Microprocessor compares digital reading to stored preset thresholds [1 mark]. 3. If temperature exceeds limit, microprocessor signals actuator to open vents/turn on fan [1 mark]. 4. Continuous closed feedback loop [1 mark].
(b) Mechanical framework, electrical power/actuators/sensors, programmable controller that operates autonomously [2 marks].

Question 2 [2210/12/M/J/23 Q8 (2023 Series)] [Total: 6 marks]

(a) An automated street lighting system turns on lights automatically when it gets dark. Explain the step-by-step operation of the system including the sensor, microprocessor, and actuator. **[4 marks]**

(b) State two advantages of automated systems over manual systems. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) 1. Light sensor (LDR) continuously measures ambient light levels [1 mark].
2. Analogue light data converted to digital signal via ADC [1 mark].
3. Microprocessor compares reading against programmed threshold [1 mark].
4. If light < threshold, signal sent to actuator/relay to switch light on; process loops continuously [1 mark].
(b) Faster response time, works 24/7 without human fatigue/error, operates in dangerous environments, consistent performance [2 marks].

Question 3 [2210/11/O/N/23 Q8 (2023 Series)] [Total: 5 marks]

(a) Explain the difference between Supervised Learning and Unsupervised Learning in Machine Learning (AI). **[3 marks]**

(b) Name two components of an Expert System and state their purpose. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) Supervised Learning: Model is trained using labeled data (input-output pairs provided) [1.5 marks].
Unsupervised Learning: Model is given unlabeled data and must discover hidden patterns/clusters on its own [1.5 marks].
(b) Knowledge Base: Stores factual knowledge and expert IF-THEN rules [1 mark].
Inference Engine: Reasoning engine that evaluates rules against user inputs to deduce conclusions [1 mark].
